

Lesson 05 Demo 01

Monitoring Azure

Objective: To create a proof of concept for monitoring the performance of virtual machines to ensure optimal operation and resource utilization

Tools required: Azure account with administrator access

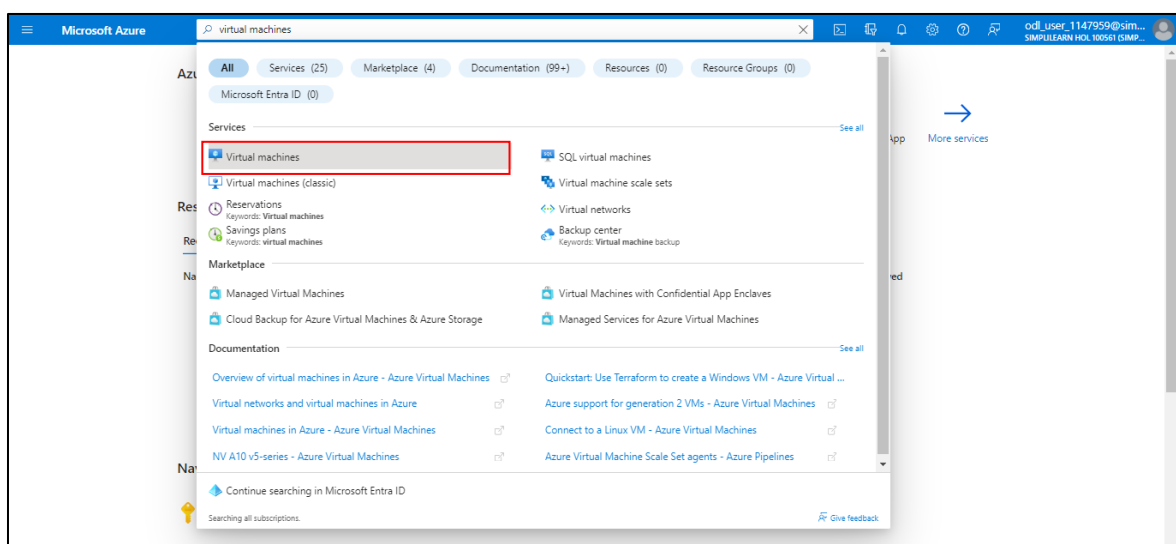
Prerequisites: None

Steps to be followed:

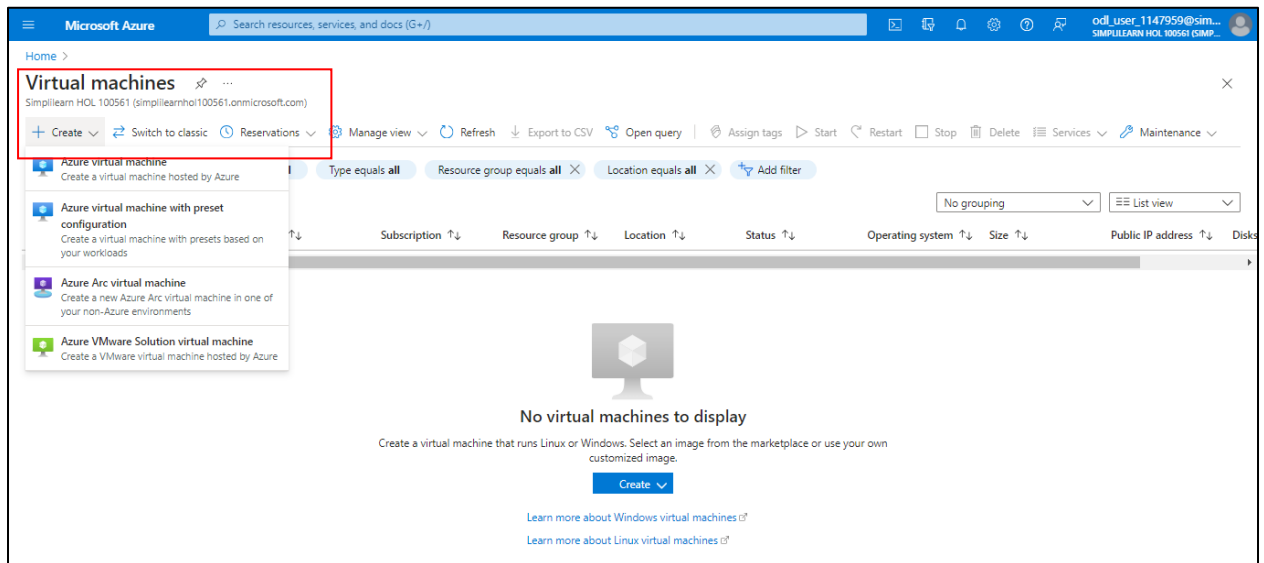
1. Deploy an Azure virtual machine
2. Create a Log Analytics workspace
3. Enable the Log Analytics virtual machine extension
4. Collect virtual machine event and performance data
5. View and query collected data

Step 1: Deploy an Azure virtual machine

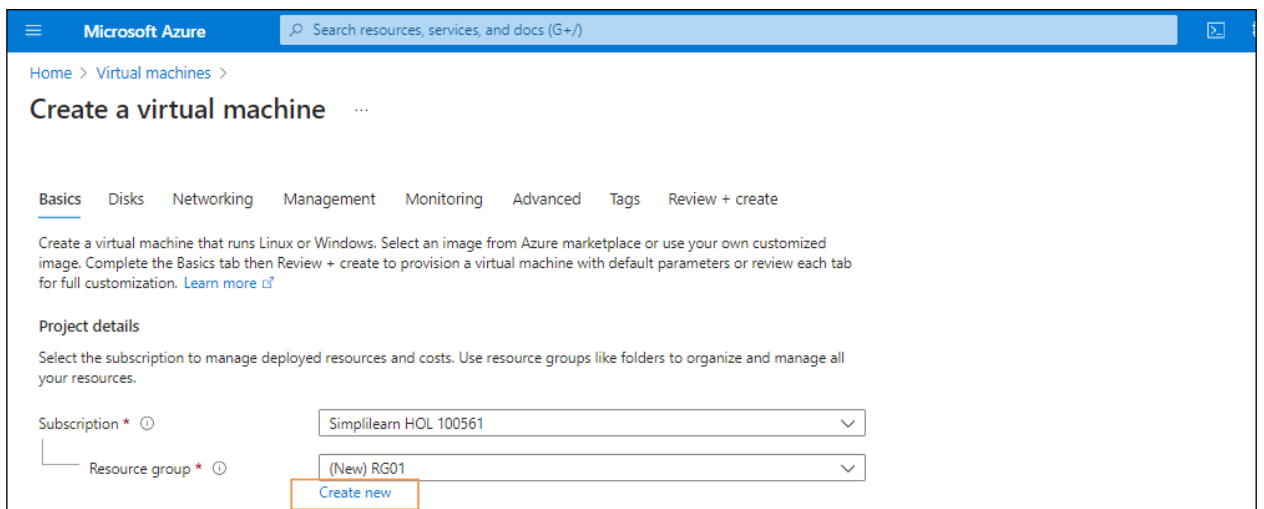
1.1 Sign in to the Azure portal; search for and select **Virtual machines**



1.2 In the **Virtual machines** page, click on **+ Create** and then select **Azure virtual machine**



1.3 In the **Create a virtual machine** page, navigate to the **Basics** tab, and under **Project details**, click on **Create new** to create a new **Resource group**



1.4 Now, under **Instance details**, enter the details as shown in the screenshot below:

Microsoft Azure

Search resources, services, and docs (G+)

Home > Virtual machines >

Create a virtual machine

Instance details

Virtual machine name *

Region *

Availability options

Security type

Image *

[See all images](#) | [Configure VM generation](#)

☒ This image is compatible with additional security features. [Click here to swap to the Trusted launch security type.](#)

VM architecture ☐ Arm64 ☒ x64

☒ Arm64 is not supported with the selected image.

Run with Azure Spot discount ☐

Size *

1.5 Under **Administrator account** and **Inbound port rules**, enter the details as shown in the screenshot below:

Microsoft Azure

Search resources, services, and docs (G+)

Home > Virtual machines >

Create a virtual machine

Administrator account

Username *

Password *

Confirm password *

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ☐ None ☒ Allow selected ports

Select inbound ports *

☒ All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

1.6 Now, click on the **Disks** tab and select **Standard HDD (locally-redundant storage)** and then click on **Next: Networking >**

Microsoft Azure Search resources, services, and docs (G+/)

Home > Virtual machines >

Create a virtual machine

Basics **Disks** Networking Management Monitoring Advanced Tags Review + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ☐

OS disk

OS disk size

OS disk type *

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM ☒

Key management

[Review + create](#) [< Previous](#) [Next : Networking >](#)

1.7 In the **Networking** tab, click on **Create new** under **Public IP**

Microsoft Azure

Search resources, services, and docs (G+)

Home > Virtual machines >

Create a virtual machine

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.

[Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network *

(new) Vm01-vnet

Create new

Subnet *

(new) default (10.0.0.0/24)

Public IP

(new) Vm01-ip

Create new

NIC network security group

None

Basic

Advanced

Public inbound ports *

None

Allow selected ports

Review + create

< Previous

Next : Management >

1.8 In the **Create public IP address panel**, select **Name** from the dropdown menu and click on **OK**

The screenshot shows the 'Create a virtual machine' page in the Microsoft Azure portal, specifically the 'Networking' tab. On the right side, a 'Create public IP address' panel is open. In this panel, the 'Name' dropdown menu is set to 'Vm01-ip', and the 'OK' button at the bottom right is highlighted with a red rectangular box. The main configuration area on the left shows various network settings: 'Virtual network' is set to '(new) Vm01-vnet', 'Subnet' is '(new) default (10.0.0.0/24)', and 'Public IP' is '(new) Vm01-ip'. Below these, there are radio button options for 'NIC network security group' (Basic is selected) and 'Public inbound ports' (Allow selected ports is selected).

1.9 Now, click on **Review + create**

This screenshot shows the same 'Create a virtual machine' page in the Microsoft Azure portal, but now the 'Review + create' button at the bottom left of the main configuration area is highlighted with a red rectangular box. The 'Networking' tab is still active, and all the network configuration details (Virtual network, Subnet, Public IP, NIC network security group, and Public inbound ports) remain the same as in the previous screenshot.

1.10 After the validation passes, click on **Create**

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machines >

Create a virtual machine

Validation passed

Basics Disks Networking Management Monitoring Advanced Tags **Review + create**

Cost given below is an estimate and not the final price. Please use [Pricing calculator](#) for all your pricing needs.

Price

1 X Standard D2s v3
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Pricing not available for this offering

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; and (b) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

You have set RDP port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go to the Networking tab.

Create < Previous Next > [Download a template for automation](#)

The following screen will appear:

Microsoft Azure Search resources, services, and docs (G+)

Home >

CreateVm-MicrosoftWindowsServer.WindowsServer-201-20231114115915 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview Inputs Outputs Template

Your deployment is complete

Deployment name: CreateVm-MicrosoftWindowsServer.WindowsServer-201-20231114115915 Start time: 14/11/2023, 12:01:18
Subscription: Simplilearn HOL 100561 Correlation ID: 4b8c03b5-c53b-4cce-a5ac-111a4b7a71bd
Resource group: RG01

Deployment details

Next steps

Setup auto-shutdown Recommended
Monitor VM health, performance and network dependencies Recommended
Run a script inside the virtual machine Recommended

[Go to resource](#) [Create another VM](#)

Give feedback
Tell us about your experience with deployment

Deployment succeeded
Deployment 'CreateVm-MicrosoftWindowsServer.WindowsServer-201-20231114115915' to resource group 'RG01' was successful.
[Go to resource](#) [Pin to dashboard](#)

Cost Management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

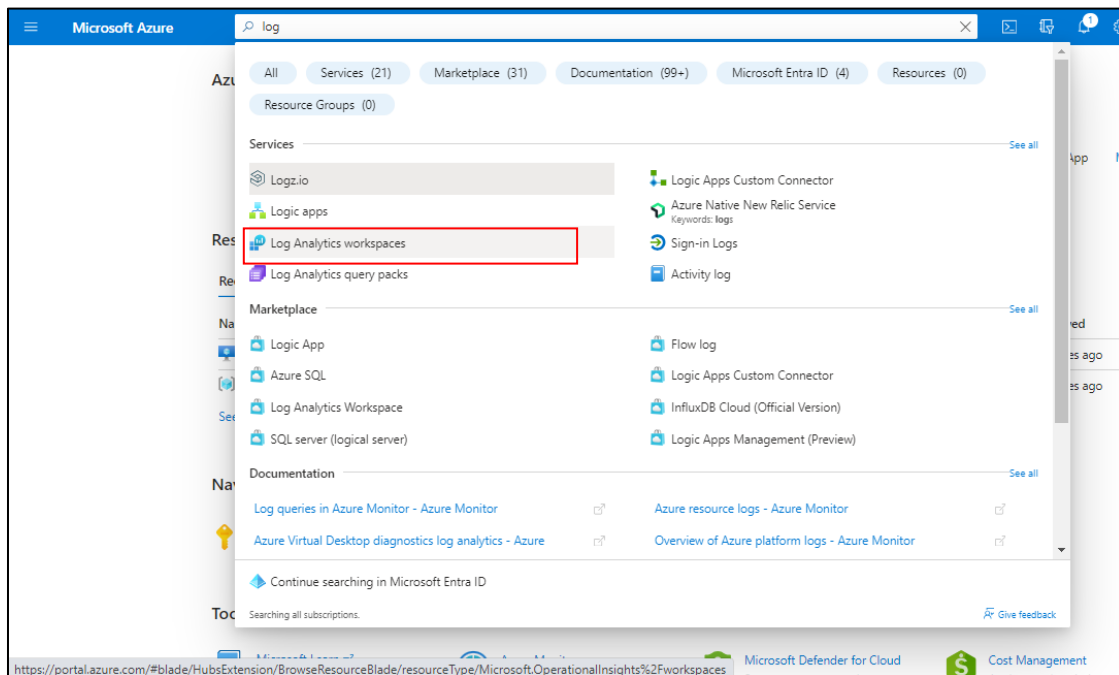
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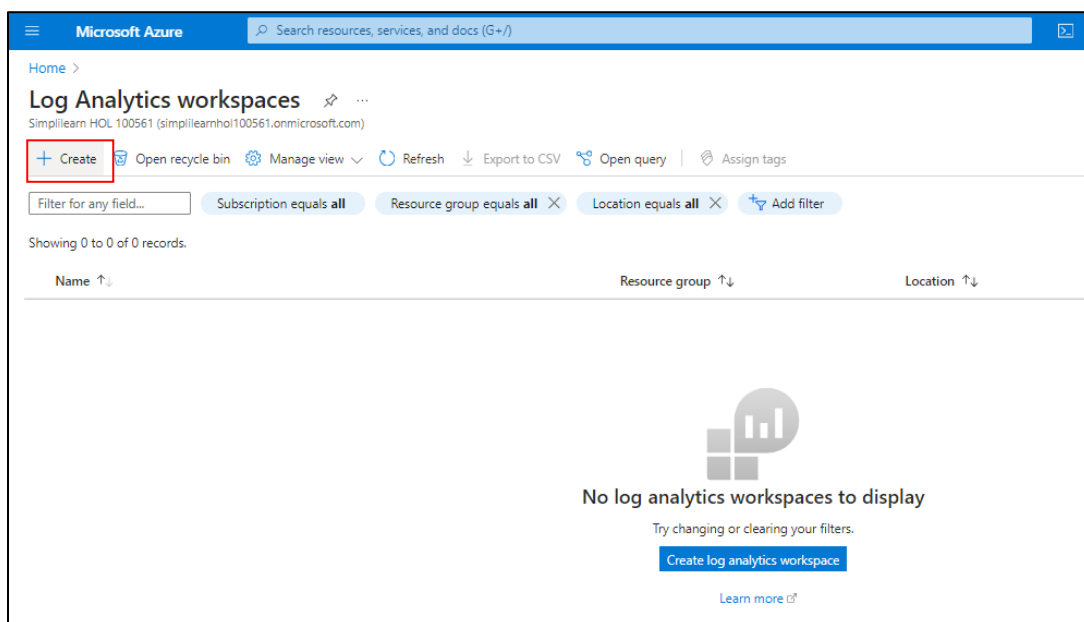
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
[Find an Azure expert >](#)

Step 2: Create a Log Analytics workspace

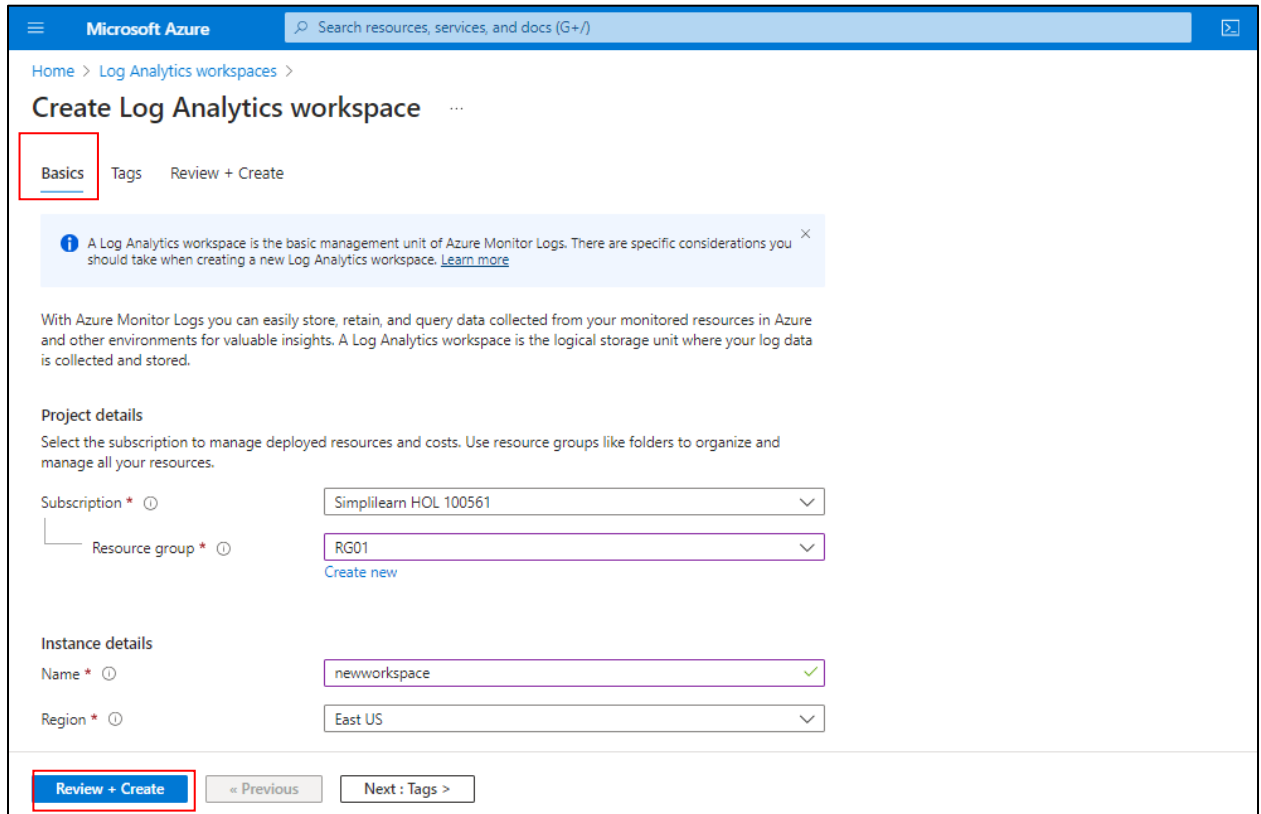
2.1 In the Azure portal, search for and select **Log Analytics workspaces**



2.2 In the **Log Analytics workspaces** page, click on **+ Create**



2.3 In the **Create Log Analytics workspace** page, click on the **Basics** tab and enter the details as shown in the screenshot below. Then, click on **Review + Create**.



Microsoft Azure Search resources, services, and docs (G+)

Home > Log Analytics workspaces >

Create Log Analytics workspace

Basics Tags Review + Create

Basics

Basics

A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance details

Name *

Region *

Review + Create « Previous Next : Tags >

2.4 After the validation passes, click on **Create**

Microsoft Azure Search resources, services, and docs (G+)

Home > Log Analytics workspaces >

Create Log Analytics workspace ...

Validation passed

Basics Tags Review + Create

Log Analytics workspace
by Microsoft

Basics

Subscription	Simplilearn HOL 100561
Resource group	RG01
Name	newworkspace
Region	East US

Pricing

Pricing tier	Pay-as-you-go (Per GB 2018)
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The cost of your workspace depends on the volume of data ingested and how long it is retained. Regional pricing details are available on the [Azure Monitor pricing page](#). You can change to a different pricing tier after the workspace is created. [Learn more](#) about Log Analytics pricing models.

Tags

Create « Previous Download a template for automation

The following screen will appear:

Microsoft Azure Search resources, services, and docs (G+)

Home >

Microsoft.LogAnalyticsOMS | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : Microsoft.LogAnalyticsOMS
Subscription : Simplilearn HOL 100561
Resource group : RG01

Start time : 11/14/2023, 12:15:55 PM
Correlation ID : 986fc4a2-a449-4a10-a602-08245291038a

Deployment details

Next steps

Go to resource

Give feedback

[Tell us about your experience with deployment](#)

Deployment succeeded
Deployment 'Microsoft.LogAnalyticsOMS' to resource group 'RG01' was successful.
[Go to resource](#) [Pin to dashboard](#)

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

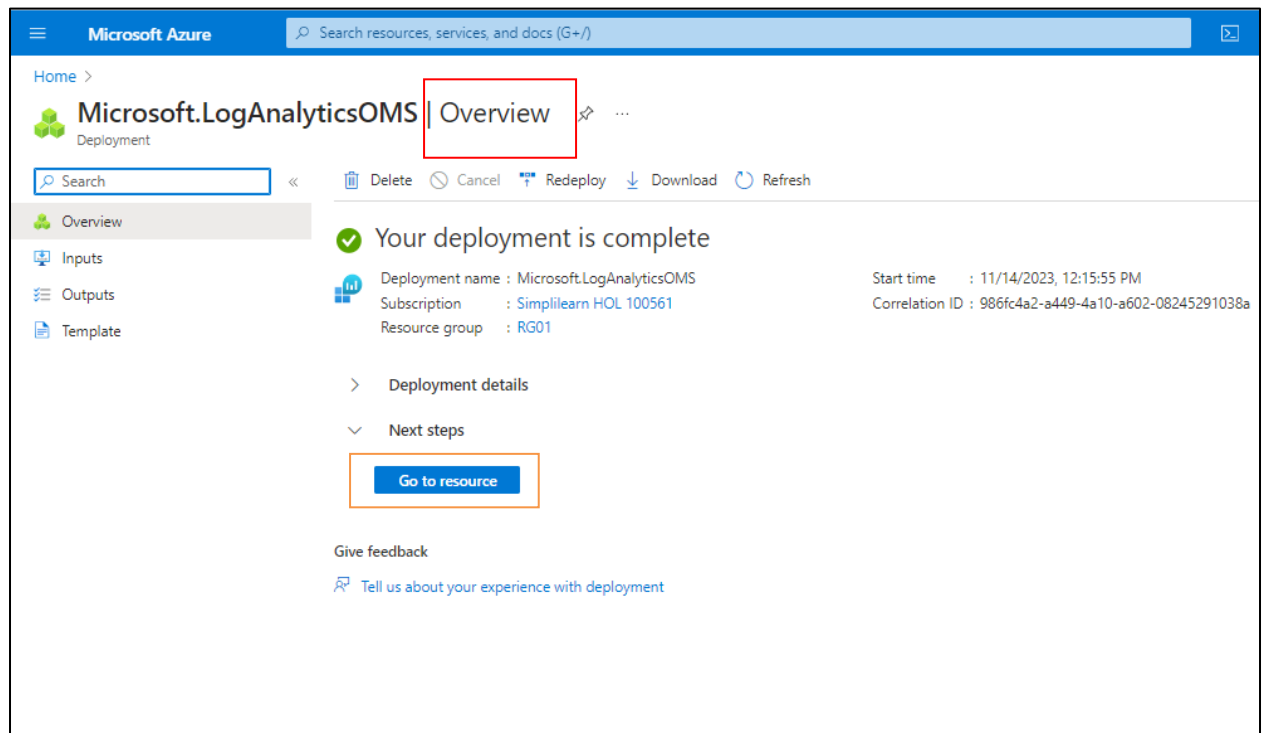
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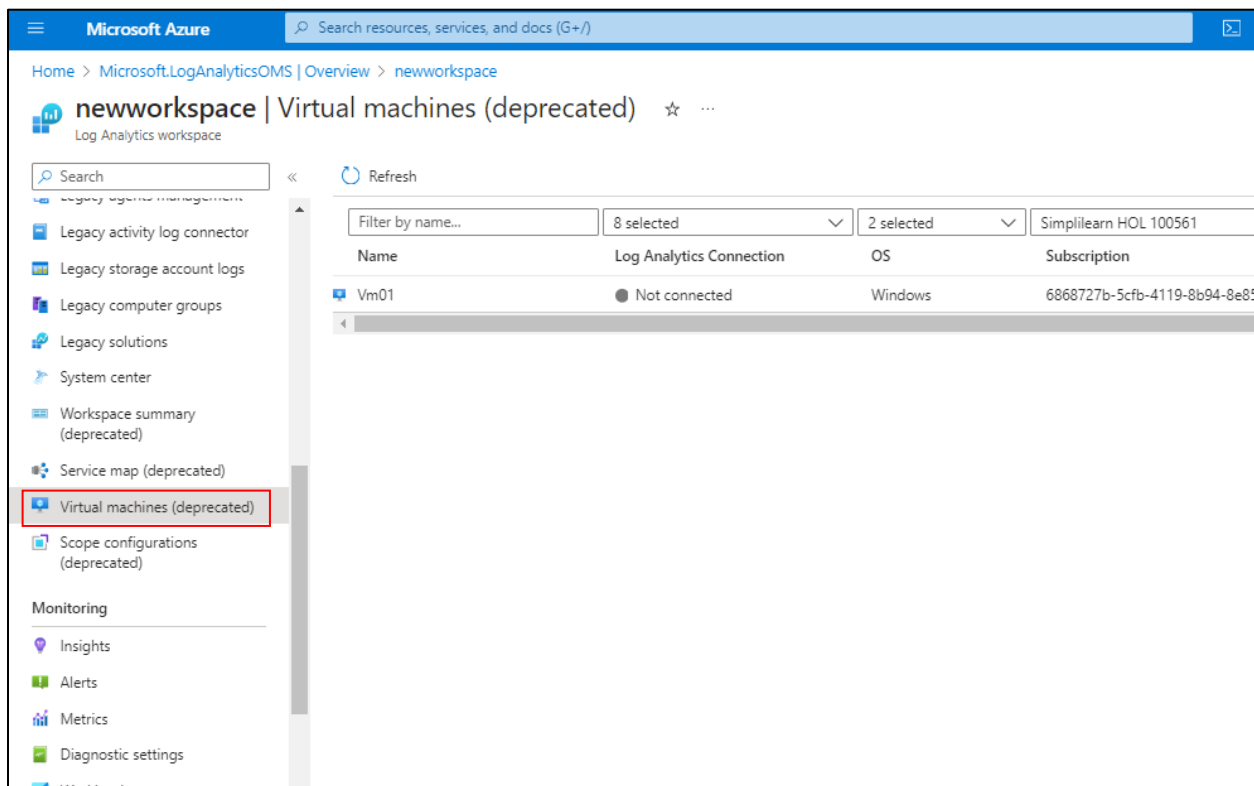
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Step 3: Enable the Log Analytics virtual machine extension

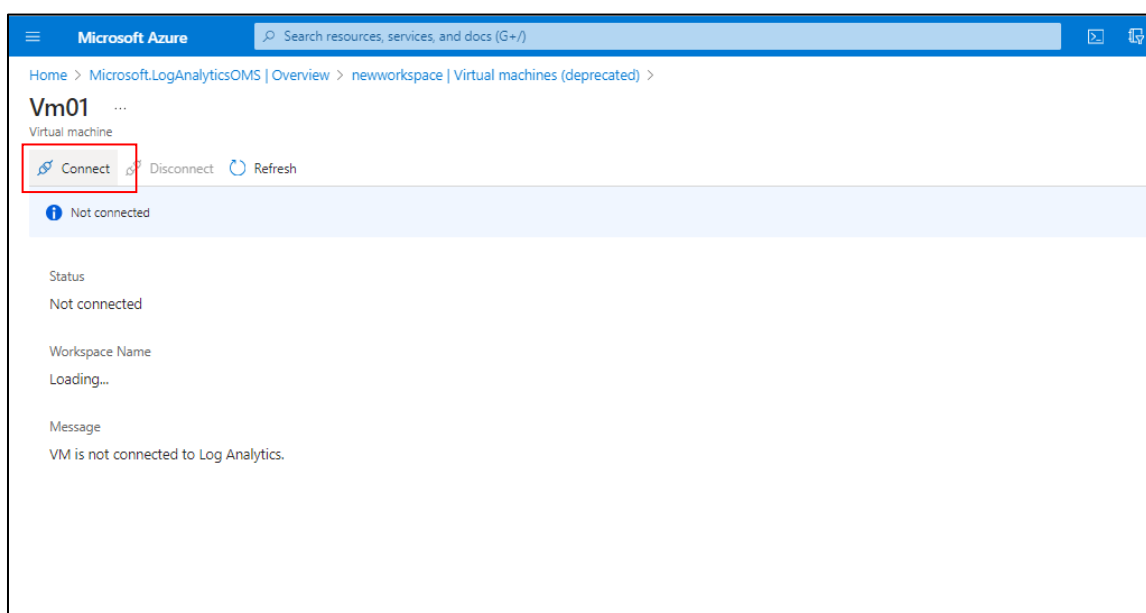
3.1 In the **Overview** page, click on **Go to resource** (same screenshot as in the previous step)



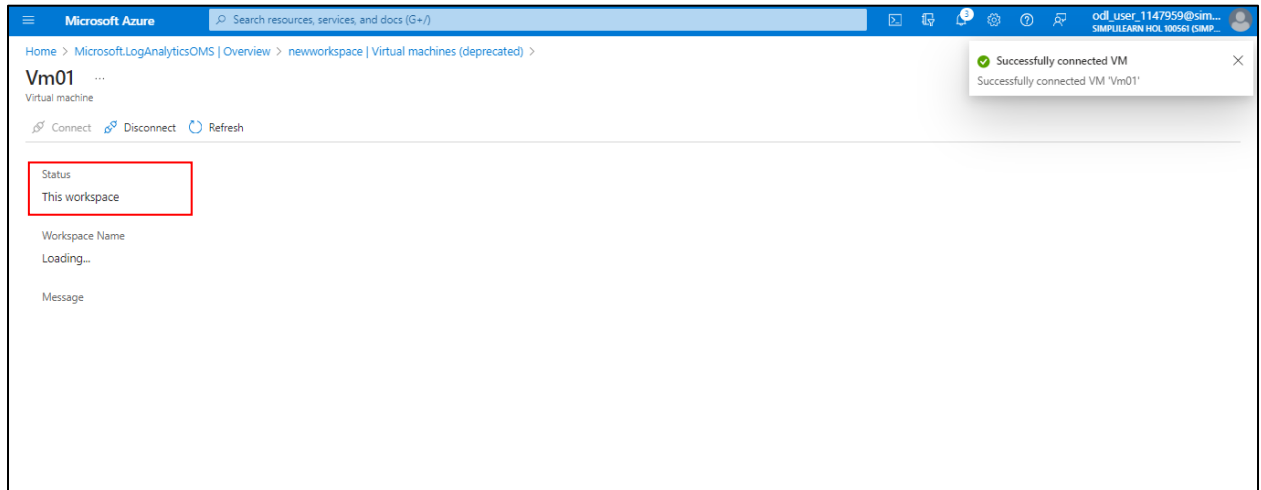
3.2 Click on **Virtual machines (deprecated)** and then select the virtual machine created earlier (**Vm01**)



3.3 In **Vm01** page, click on **Connect**

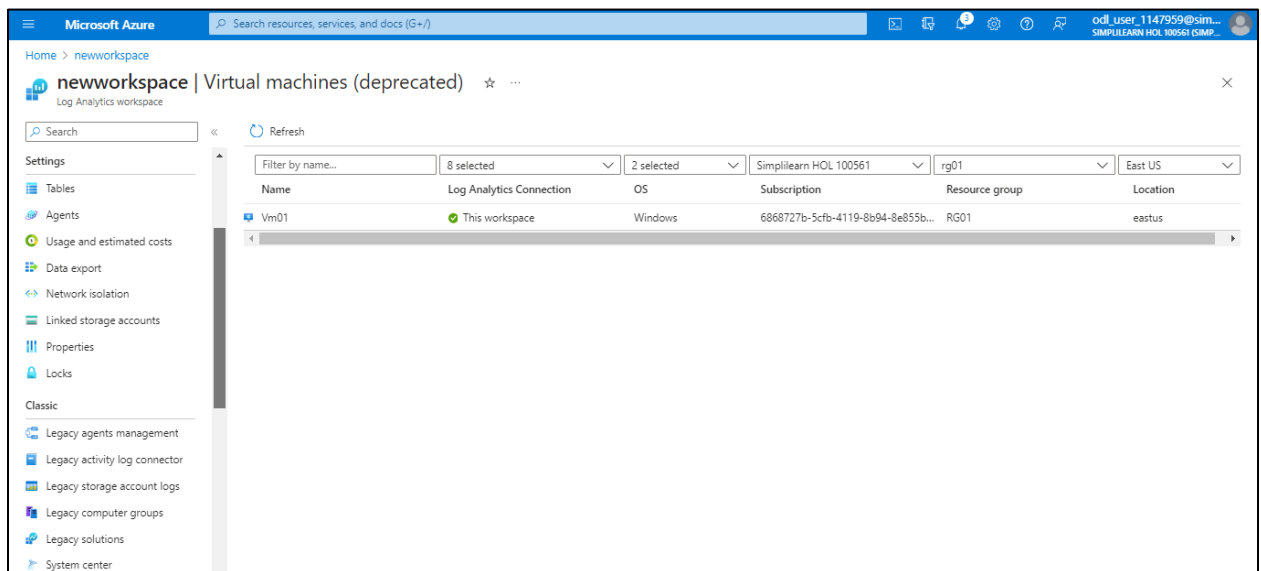


This may take a few minutes. The **Status** displayed on **Vm01** page will change from **Connecting** to **This workspace** as shown in the screenshot below:



Step 4: Collect virtual machine event and performance data

4.1 Navigate back to the Log Analytics workspace(newworkspace) created earlier



4.2 Under **Classic** section, select **Legacy agents management** and then click on **+ Add windows event log**

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo and a search bar. The breadcrumb trail indicates the user is in the 'newworkspace' workspace. The left-hand navigation pane is expanded to the 'Classic' section, where 'Legacy agents management' is selected. The main content area displays the 'Legacy agents management' page for the 'newworkspace' workspace. A warning message at the top states: 'The Log Analytics agents won't be supported as of August 31, 2024. Plan to migrate to Azure Monitor Agent prior to this date. If you've already installed Azure Monitor Agent, make sure to create and associate [data collection rules](#) to the agents.' Below the warning, there is a section titled 'Collect Windows event log data from standard logs, like System and Application, or add custom logs created by applications you need to monitor. [Learn more](#)'. A button labeled '+ Add windows event log' is highlighted with a red box. Below this button is a 'Filter event logs' input field. A table with columns 'Log name', 'Error', 'Warning', and 'Information' is visible. At the bottom of the page, there are 'Apply' and 'Discard changes' buttons.

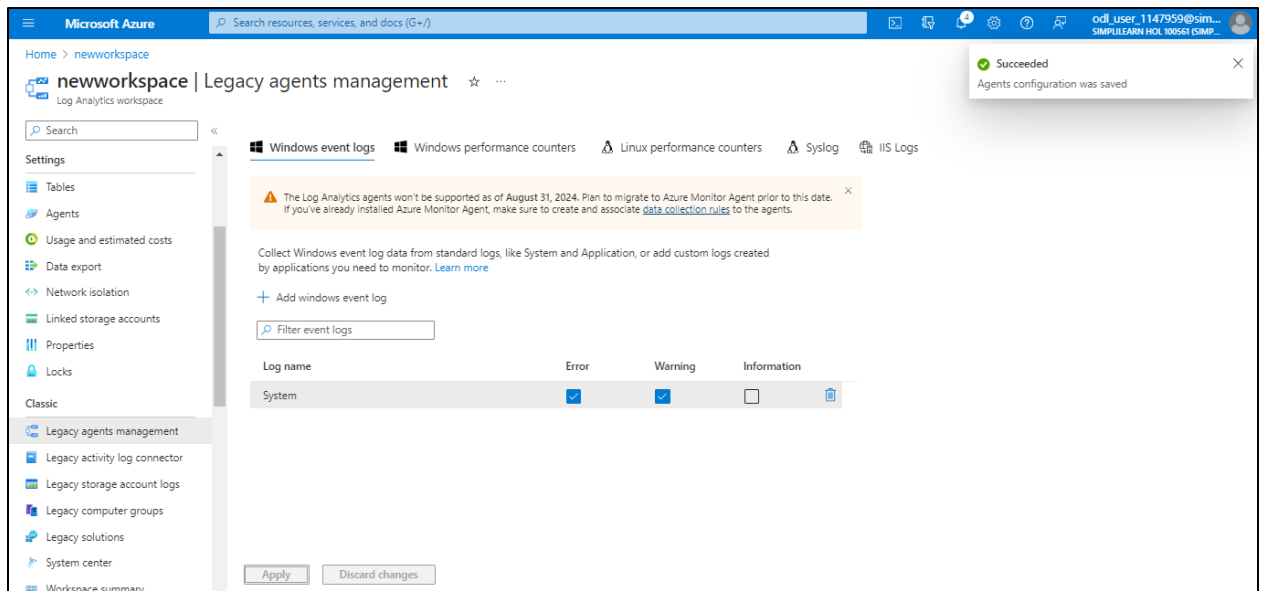
4.3 Under **Log name** select **System**, uncheck the **Information** box and then click on **Apply**

The screenshot shows the Microsoft Azure portal interface for a Log Analytics workspace named 'newworkspace'. The 'Legacy agents management' page is open, specifically the 'Windows event logs' tab. A navigation pane on the left lists various settings and classic tools. A warning message at the top indicates that Log Analytics agents will no longer be supported after August 31, 2024, and suggests migrating to the Azure Monitor Agent. Below this, instructions are provided on how to collect Windows event log data. A table of event logs is displayed, with columns for 'Log name', 'Error', 'Warning', and 'Information'. The 'System' log is selected, and the 'Information' checkbox is unchecked. The 'Apply' button is highlighted at the bottom.

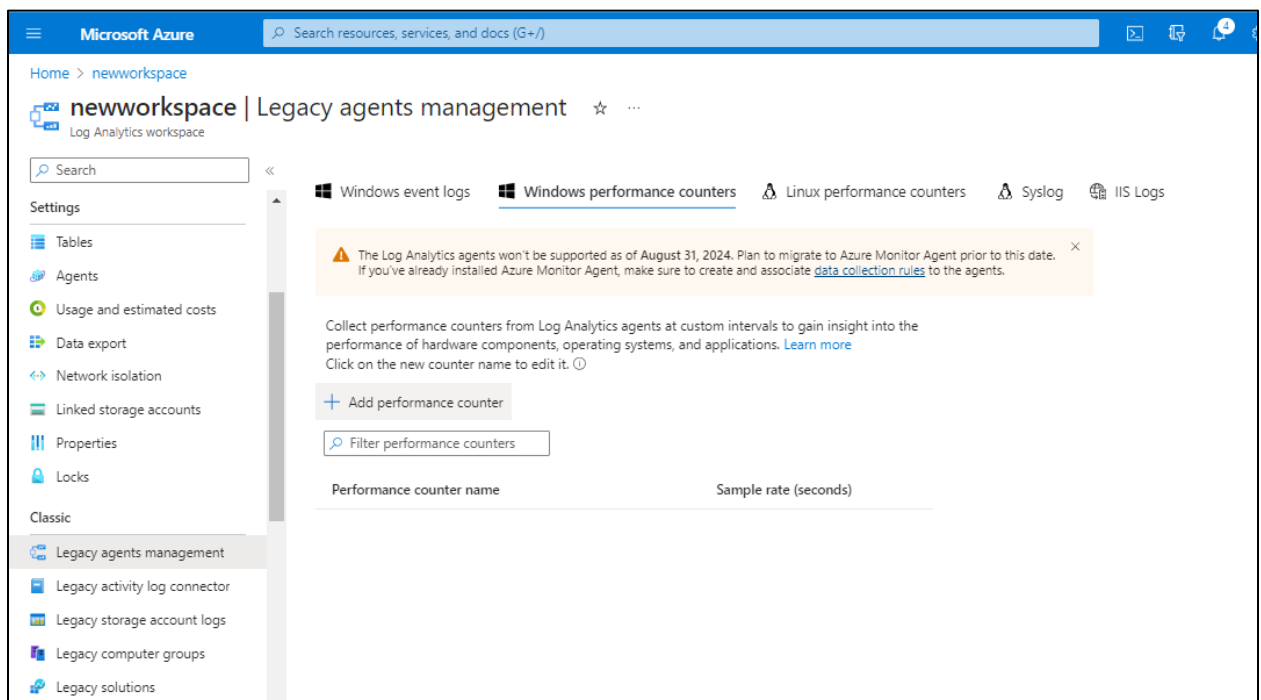
Log name	Error	Warning	Information
System	☒	☒	☐

Note: You can also add events like **Hardware events** or **Key Management service** per the requirement.

The following screenshot will appear:



4.4 Click on **Windows performance counters** and select **+ Add performance counter**



4.5 Review the listing of available performance counters. Then, add the following ones and click on **Apply**

Process(*)\% Processor Time

Event Tracing for Windows(*)\Total Memory Usage — Non-Paged Pool

Event Tracing for Windows(*)\Total Memory Usage — Paged Pool

Microsoft Azure Search resources, services, and docs (G+)

Home > newworkspace

newworkspace | Legacy agents management ☆ ...

Log Analytics workspace

Search

Settings

- Tables
- Agents
- Usage and estimated costs
- Data export
- Network isolation
- Linked storage accounts
- Properties
- Locks
- Classic
- Legacy agents management
- Legacy activity log connector
- Legacy storage account logs
- Legacy computer groups
- Legacy solutions
- System center
- Workspace summary

Windows event logs Windows performance counters Linux performance counters Syslog IIS Logs

The Log Analytics agents won't be supported as of August 31, 2024. Plan to migrate to Azure Monitor Agent prior to this date. If you've already installed Azure Monitor Agent, make sure to create and associate [data collection rules](#) to the agents.

Collect performance counters from Log Analytics agents at custom intervals to gain insight into the performance of hardware components, operating systems, and applications. [Learn more](#)

Click on the new counter name to edit it. ⓘ

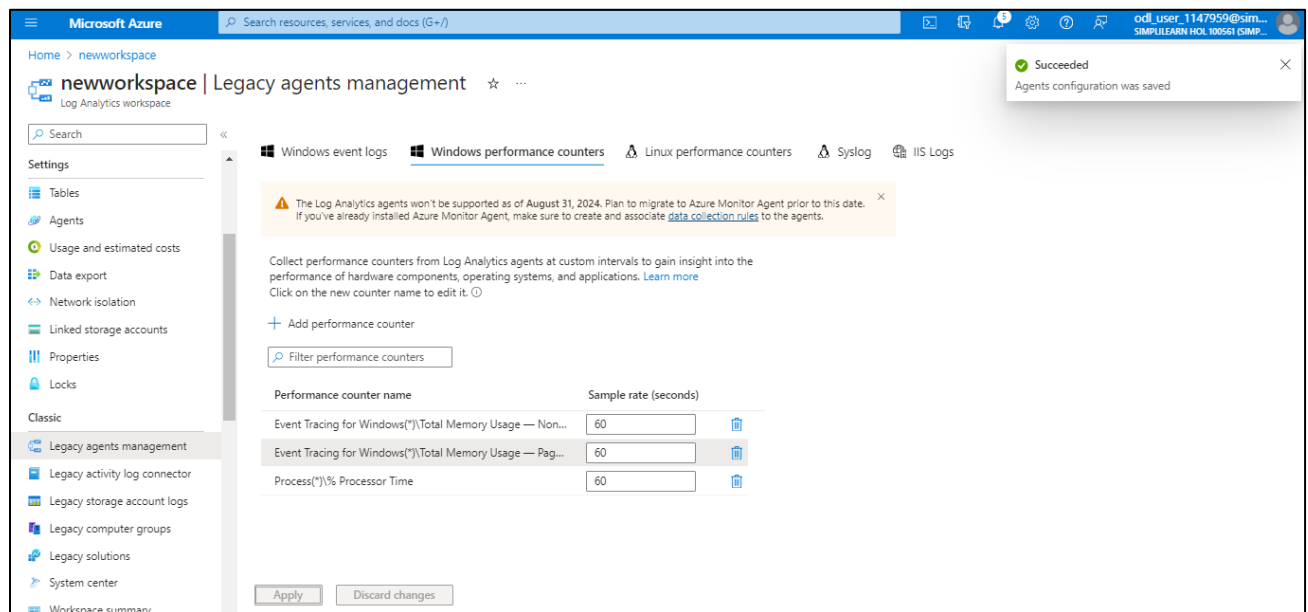
+ Add performance counter

Filter performance counters

Performance counter name	Sample rate (seconds)
Event Tracing for Windows(*)\Total Memory Usage — P...	60
Event Tracing for Windows(*)\Total Memory Usage — ...	60
Process(*)\% Processor Time	60

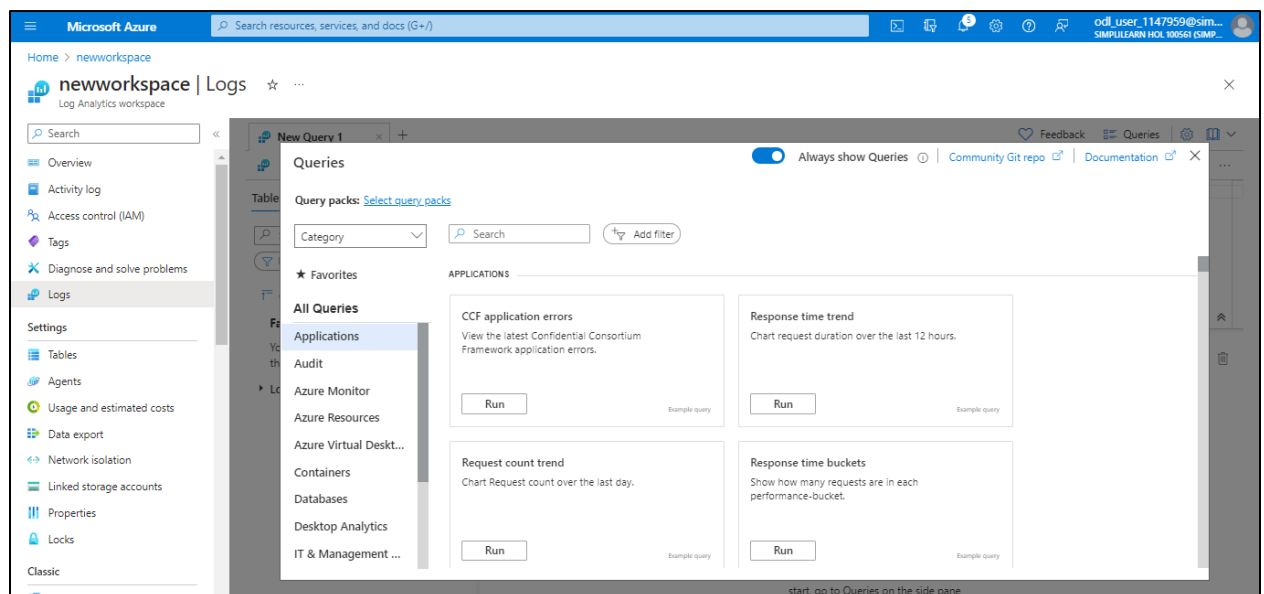
Apply Discard changes

The following screenshot will appear:

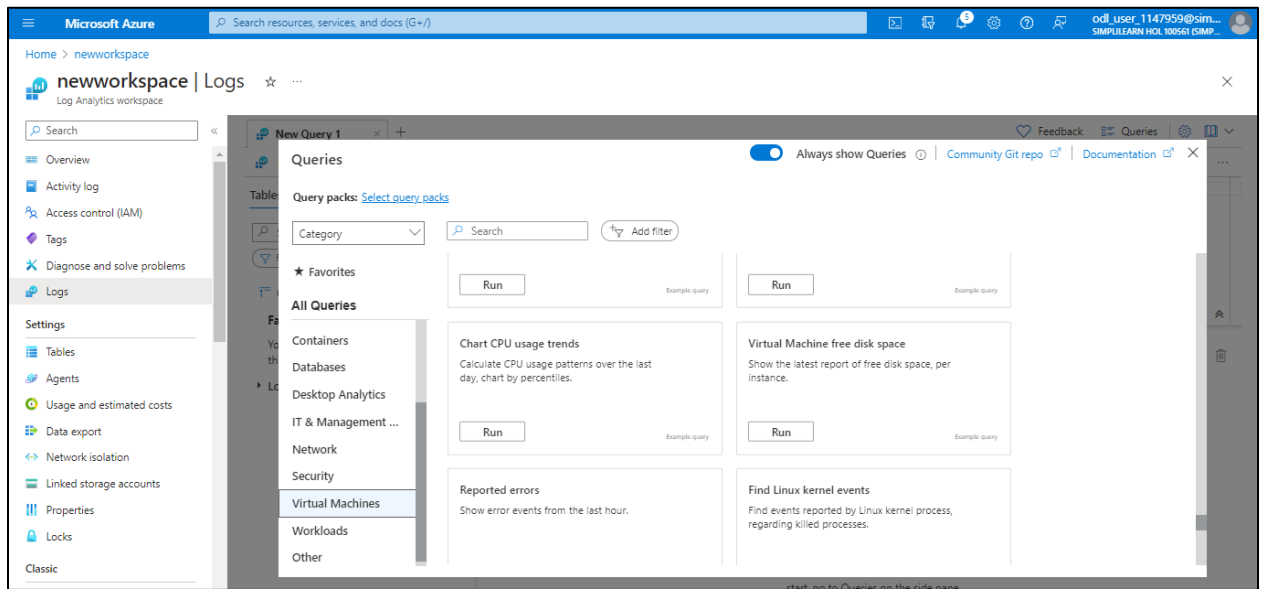


Step 5: View and query collected data

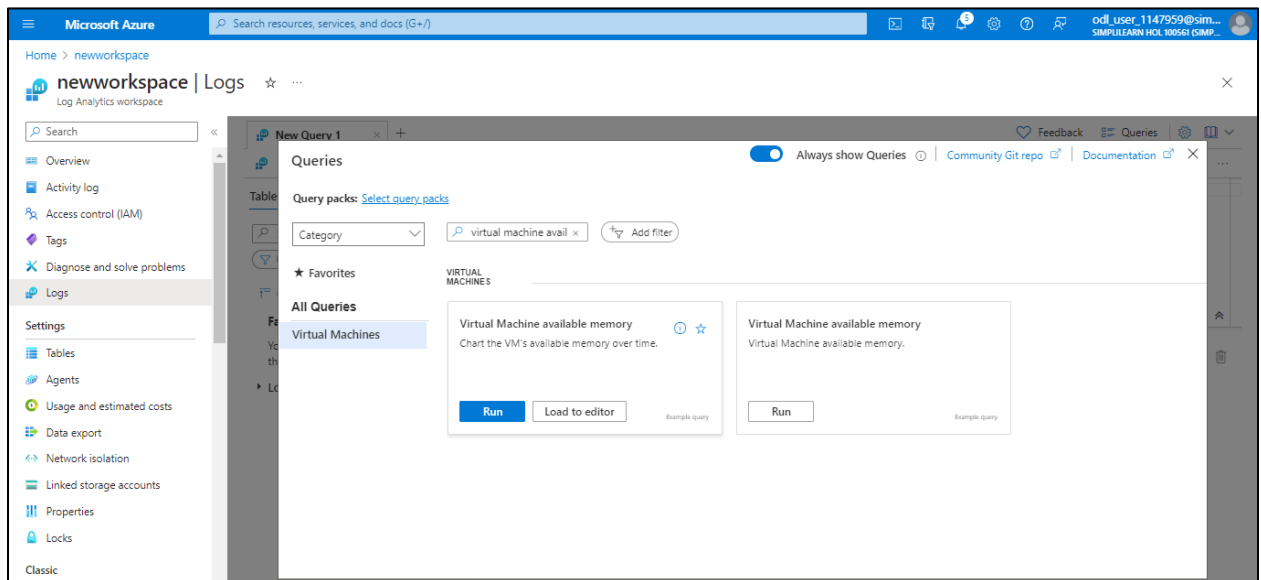
5.1 Navigate back to the Log Analytics workspace (**newworkspace**) created earlier and select **Logs** in the left panel



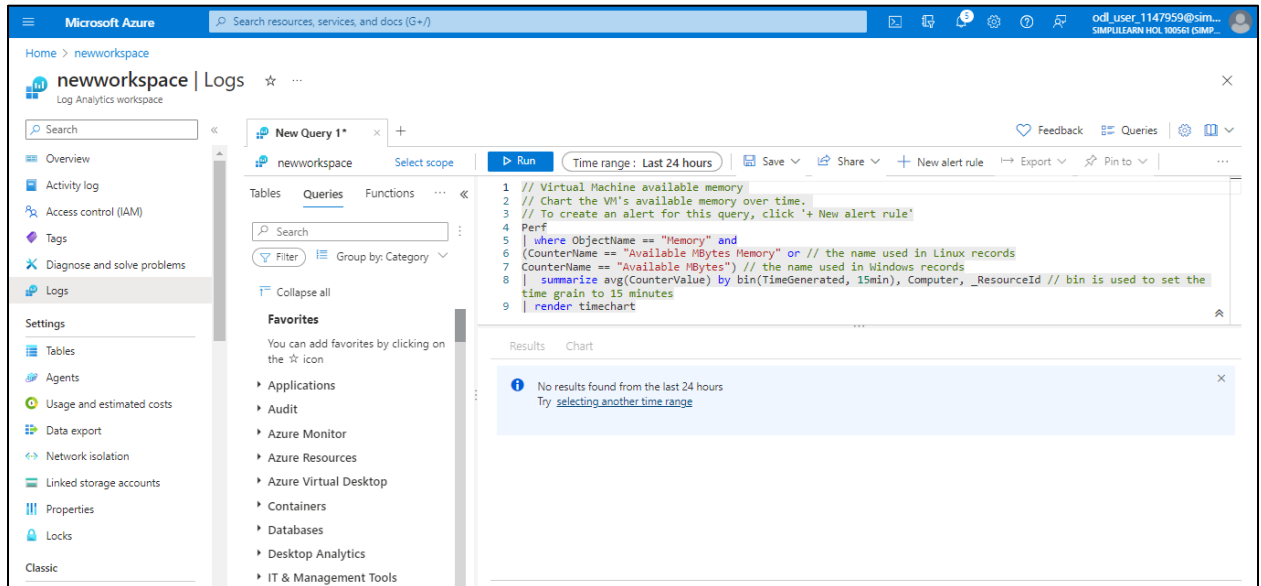
5.2 In the **Logs** page, select **Virtual Machines** under **All Queries**



5.3 Now, review the list of predefined queries, identify the one you want to test, and click the corresponding **Run** button



The query will automatically open in a **New Query** tab as shown in the screenshot below:



By following these steps, you have successfully created a proof of concept for monitoring the use of the Log Analytics workspace to configure data sources and query logs.